Steerable Personal AI Assistants

Multimodal correction and negotiated initiative in everyday life

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Research vision and motivation

“Put that on the shelf — no, the other one.” A person in the kitchen glances at a jar, speaks, and changes their mind. Their words, gaze and correction are part of one continuing conversation. I want a home assistant that can follow it: one people can steer naturally while they are doing something else.

My long-term vision is a home Jarvis that sees and hears what matters, remembers useful preferences and supports everyday activities. It might speak, preview a rearrangement, or quietly complete an agreed task. The research starting point is the human side of that capability: how an assistant understands a correction, how much initiative it should take, and when it should stop.

Research context and opportunity

Gaze and speech can offer complementary cues about a person’s focus and meaning. Khan and colleagues [1, 2] examine their combination, including the challenge of interpreting natural gaze and speech together. I want to extend this question to an ongoing task: what should an assistant do when a person changes a reference or instruction while the action is underway?

In Syiem and colleagues’ AR study [3], the adaptive agent did not significantly improve task performance over unassisted AR; people’s preferences, trust and understanding affected their use of its assistance. Research on AI decision support also shows that interaction after an error can affect trust [4]. These findings motivate my question: even when an assistant makes useful predictions, can people readily inspect, redirect and repair its actions?

The question sits at the intersection of Professor Velloso’s work on multimodal interaction and human-centred AI. His research on AI proxies [5] and transparency in multi-agent interfaces [6] adds a useful perspective on how people understand and retain a role in AI-mediated action. These studies address different settings; I want to investigate these issues in everyday home tasks.

Research questions

How can people steer a personal assistant through speech, gaze and correction as their intentions change? When should an assistant act, preview its interpretation or ask for confirmation? How can it learn from corrections while letting people inspect and change the boundaries of its initiative?

Research approach and contribution

I would begin with a Wizard-of-Oz study in a simulated home: a person tidies and arranges objects while a hidden human operates the assistant. I want to observe how people express a change of mind, use gaze and speech together, and experience help or interruption. This initial study would inform a bounded, reversible prototype rather than assume an interaction style in advance.

I would then compare alternative interactions: a spatial preview before action, action with an accessible review and undo, and a more automatic condition. Across the same task, I would examine task completion, unwanted interruptions, error detection and correction time, and perceived control. Interviews would help explain why an intervention felt helpful in one moment and intrusive in another.

A later repeated-session study could examine how preferences and permissions evolve. Corrections may offer clues to a person's boundaries, while consequential actions would require appropriate confirmation. I hope to contribute interaction knowledge about negotiating initiative with an assistant that remains responsive as people change their minds.

Preparation

Before my Master of Computer Science, I worked as an architect; architecture taught me to think in rooms, objects and how people move between them. For Signal Box, I helped design and test an interactive prototype, revising feedback and information sharing after user testing. My Game Fellows prototype explores a rule-based companion; an evaluation comparing play with and without it is planned. In my explainable image-detection capstone, I independently reconstructed 900 fusion decisions to examine where the system went wrong. I would love to develop this research with Professor Velloso.

References

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